

# On the Performance of CNNs and Vision Transformers under Resource-Constrained Training: A CIFAR-10 Study with Diffusion Models

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**Abstract**—This paper presents an empirical study comparing Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and Diffusion Models on the CIFAR-10 dataset under strict resource-constrained conditions. While recent advances in computer vision rely heavily on large-scale datasets and GPU-intensive training, this work focuses on a practical setting where only CPU-based training is available.

We implement all models from scratch using PyTorch and evaluate their performance in terms of classification accuracy, training dynamics, and generative capability. Experimental results show that CNN achieves 66.86% accuracy, significantly outperforming the Vision Transformer at 49.81%, highlighting the importance of spatial inductive bias in low-data and low-compute regimes. Additionally, a diffusion model is trained for image generation, demonstrating stable noise learning but limited visual quality due to restricted training steps.

This study provides insights into the limitations of modern architectures when computational resources are constrained and emphasizes the importance of model efficiency for real-world deployment scenarios.

**Index Terms**—Convolutional Neural Networks, Vision Transformers, Diffusion Models, CIFAR-10, Resource-Constrained Learning, CPU Training

## I. INTRODUCTION

Deep learning has significantly advanced computer vision tasks, with Convolutional Neural Networks (CNNs) dominating image classification due to their strong inductive biases. More recently, Vision Transformers (ViTs) have demonstrated competitive performance by leveraging self-attention mechanisms. In parallel, diffusion models have emerged as powerful generative frameworks capable of producing high-quality images.

However, most existing studies assume access to large datasets and high-performance GPUs. This creates a gap between theoretical performance and real-world applicability.

This work addresses this gap by evaluating CNNs, Vision Transformers, and Diffusion Models under CPU-only constraints using the CIFAR-10 dataset.

Unlike prior work that evaluates models in high-performance environments, this study explicitly focuses on CPU-only training. This constraint reflects realistic scenarios faced by independent researchers and low-resource systems, where access to GPUs is limited. By analyzing the behavior of the model under such conditions, this work aims to provide

practical guidance on architecture selection beyond idealized benchmarks.

## II. CONTRIBUTIONS

The main contributions of this work are:

- Implementation of CNN, Vision Transformer, and Diffusion models from scratch using PyTorch
- Empirical evaluation under CPU-only constraints, reflecting real-world limitations
- Comparative analysis of classification accuracy and training dynamics
- Investigation of generative performance using a simplified diffusion model
- Insights into the efficiency and practicality of modern architectures in low-resource environments

## III. RELATED WORK

CNN architectures such as AlexNet and ResNet have demonstrated strong performance in image classification by leveraging spatial hierarchies.

Vision Transformers apply transformer architectures to image patches and achieve state-of-the-art performance on large-scale datasets, but require extensive training data.

Diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPM), generate images by iteratively denoising random noise.

This work focuses on comparing these models under constrained computational environments.

## IV. METHODOLOGY

### A. Dataset

The CIFAR-10 dataset contains 60,000 images of size 32x32 across 10 classes, split into 50,000 training and 10,000 test samples.

### B. CNN

The CNN model consists of:

- Convolutional layers with ReLU activation
- Max-pooling layers
- Fully connected classification layers

### C. Vision Transformer

The Vision Transformer includes:

- Patch embedding
- Positional embeddings
- Transformer encoder layers with self-attention

### D. Diffusion Model

A simplified UNet-based diffusion model is implemented:

- Linear beta noise schedule
- Noise prediction objective using MSE loss
- Iterative denoising process

### E. Training Setup

- Device: CPU only
- Optimizer: Adam
- Loss:
  - CNN / ViT: Cross-Entropy
  - Diffusion: Mean Squared Error
- Limited epochs due to computational constraints

## V. EXPERIMENTS

The experiments evaluate:

- Classification performance (CNN vs ViT)
- Training stability using loss curves
- Generative quality of diffusion model

## VI. RESULTS

### A. Classification Performance

| Model              | Accuracy (%) |
|--------------------|--------------|
| CNN                | 66.86        |
| Vision Transformer | 49.81        |

TABLE I

CLASSIFICATION PERFORMANCE COMPARISON ON CIFAR-10

CNN significantly outperforms the Vision Transformer under limited training conditions.

### B. Training Loss

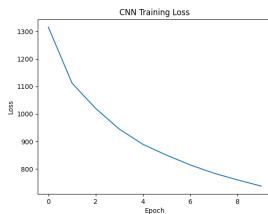


Fig. 1. CNN training loss showing stable convergence

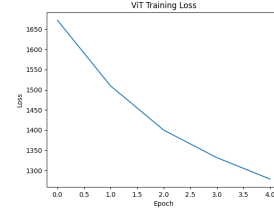


Fig. 2. ViT training loss showing stable convergence

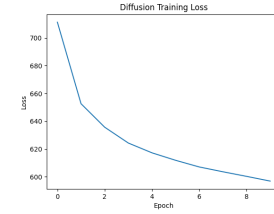


Fig. 3. Diffusion training loss showing stable convergence

### C. Generated Samples

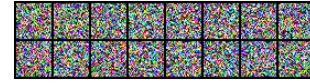


Fig. 4. Diffusion-generated images

Generated samples remain noisy due to limited training duration.

## VII. COMPUTATIONAL EFFICIENCY

In addition to accuracy, computational efficiency is a critical factor in resource-constrained environments. Table II summarizes approximate model complexity and training behavior.

| Model           | Parameters | Training Cost |
|-----------------|------------|---------------|
| CNN             | Low        | Fast          |
| ViT             | Moderate   | Slow          |
| Diffusion Model | High       | Very Slow     |

TABLE II

RELATIVE COMPARISON OF MODEL COMPLEXITY AND TRAINING COST UNDER CPU CONSTRAINTS

The Vision Transformer exhibits higher computational overhead due to self-attention operations, while the diffusion model incurs the highest cost due to iterative noise modeling across timesteps.

## VIII. DISCUSSION

The results highlight the importance of inductive bias and computational efficiency in practical machine learning settings. CNNs demonstrate strong performance due to their inherent spatial structure, allowing effective learning even with limited data and compute.

In contrast, Vision Transformers rely heavily on data scale and training duration to learn meaningful representations,

which limits their effectiveness in constrained environments. The observed performance gap reinforces the dependency of transformer-based models on large-scale training regimes.

Diffusion models, while theoretically powerful, require extensive training and computational resources to generate high-quality images. Under limited training conditions, the model primarily learns noise distributions without fully capturing semantic structure.

Overall, the findings suggest that while modern architectures offer flexibility and scalability, traditional approaches such as CNNs remain highly competitive when resources are limited.

#### IX. LIMITATIONS

- CPU-only training
- Limited epochs
- No extensive hyperparameter tuning
- Diffusion model not fully converged

#### X. CONCLUSION

This study demonstrates that CNNs remain strong baselines in low-resource settings, while modern architectures such as Vision Transformers and diffusion models require significant computational resources to achieve optimal performance.

Future work will explore scaling these models with GPU acceleration and incorporating hybrid architectures to bridge the performance gap observed in constrained environments.

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